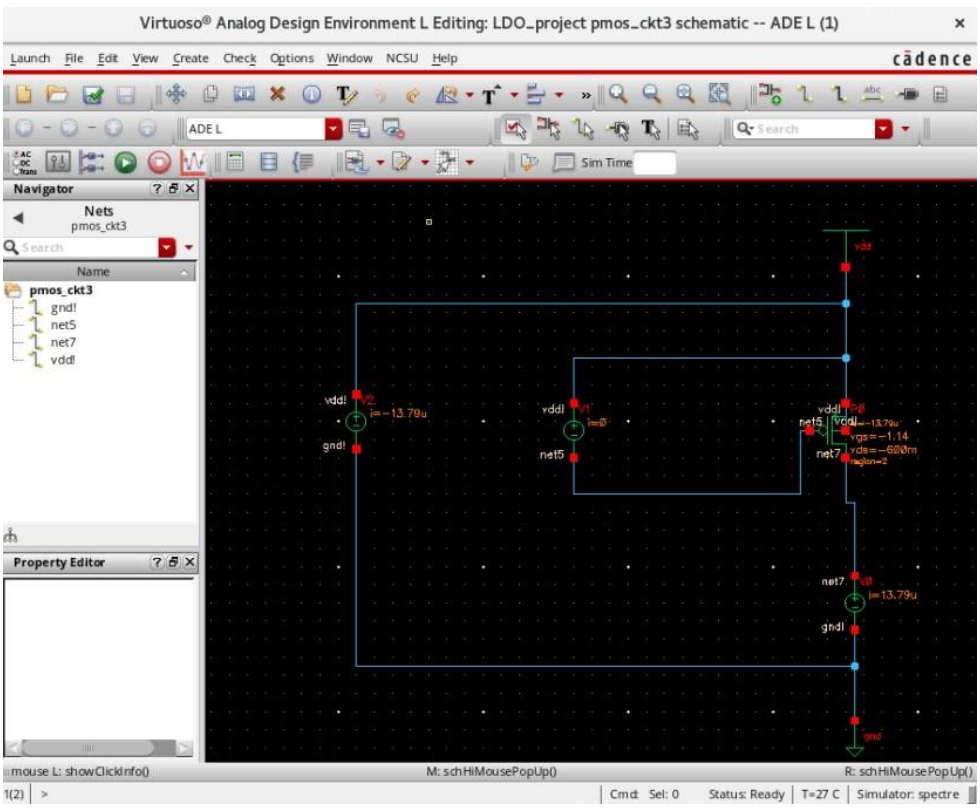
LDO Design

Requirements:

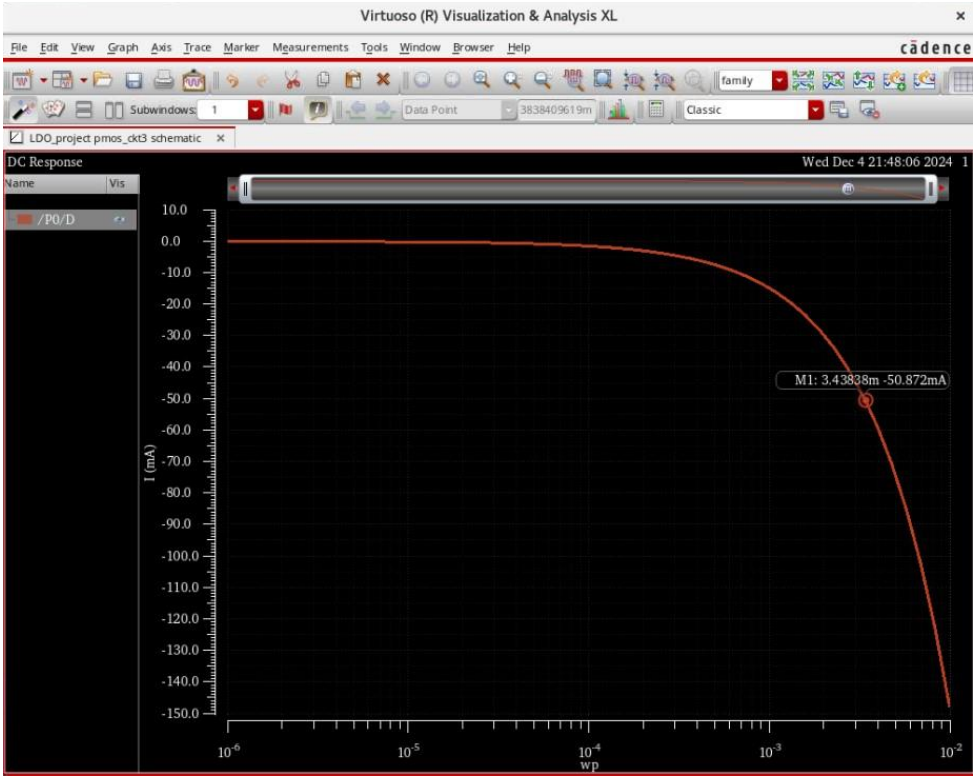
- Voltage supply 2.2V, CMOS TSMC 0.25µm Process
- VREF = 1V
- Output capacitor = 50 pF
- Delivers 1 mA~50 mA current (Depending on the load requirement)
- Output Regulated Voltage of 1.6V , VRipple < 5%
- PDC < 0.3mW Task 1:

Driver Stage:

a) **Schematic:**



b) **Wp V/S Ids**



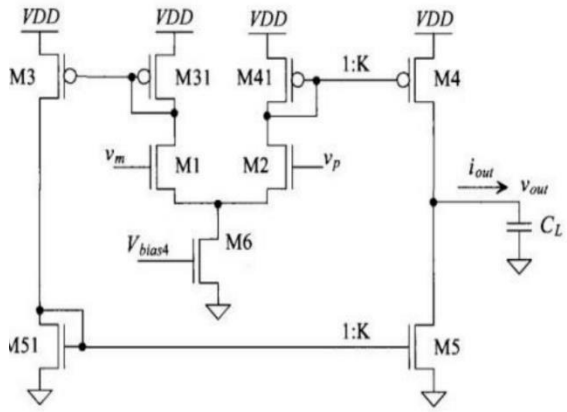
c) Input Capacitance:

Input Capacitance was measured approximately as ~ 10pF

Task 2:

2) Amplifier circuit:

DC Analysis: Calculations:



$P_{DC} = I_D \times V_{DD}$

$I_D = 126\,\mu$

$V_{DD} = 2.2\,V$

$\therefore P_{DC} = 126\mu \times 2.2V$,

$P_{DC} = 0.27\,mW$

$P_{DC} < 0.3\,mV$

$M_7 = \frac{1}{2}\,\mu n C_{ox} \left(\frac{W}{L}\right) V_{ov}^2 = I$,

$1\mu A = \frac{1}{2} \times 2.7475 \times 10^{-4} \times (W/L)\ (0.1)^2$

$\frac{W}{L} \Rightarrow 0.72 \Rightarrow L = 600\,nm$

$W = 432\,nm \approx 450\,nm$

$M_6 \Rightarrow I = 24\,\mu A \Rightarrow W = 24 \times 450\,nm$

$L = 600\,nm$, $W = 10.8\,\mu m$

$M_1, M_2 :- I = 12\,\mu A \Rightarrow W = 10.8/2$

$L = 600\,nm$ $W = 5.4\,\mu m$

$M_{31}, M_{41}, = I = 12\,\mu A = \frac{1}{2} \times 0.9576 \times 10^{-4} \times (W/L) \times 0.1$

$\frac{W}{L} = 25.06 \Rightarrow L = 600\,nm, W = 15.036\,\mu m$

$M_{S1} = 10\,\mu A = \frac{1}{2} \times 2.7475 \times 10^{-4} \times \frac{W}{L} \times (0.1)^2$

$W/L = 7.24$, $L = 600\,\mu m$, $W = 4.36\,\mu m$

M_5 :- $I = 100\,\mu A \Rightarrow W = 43.6\,\mu m$, $L = 600\,nm$

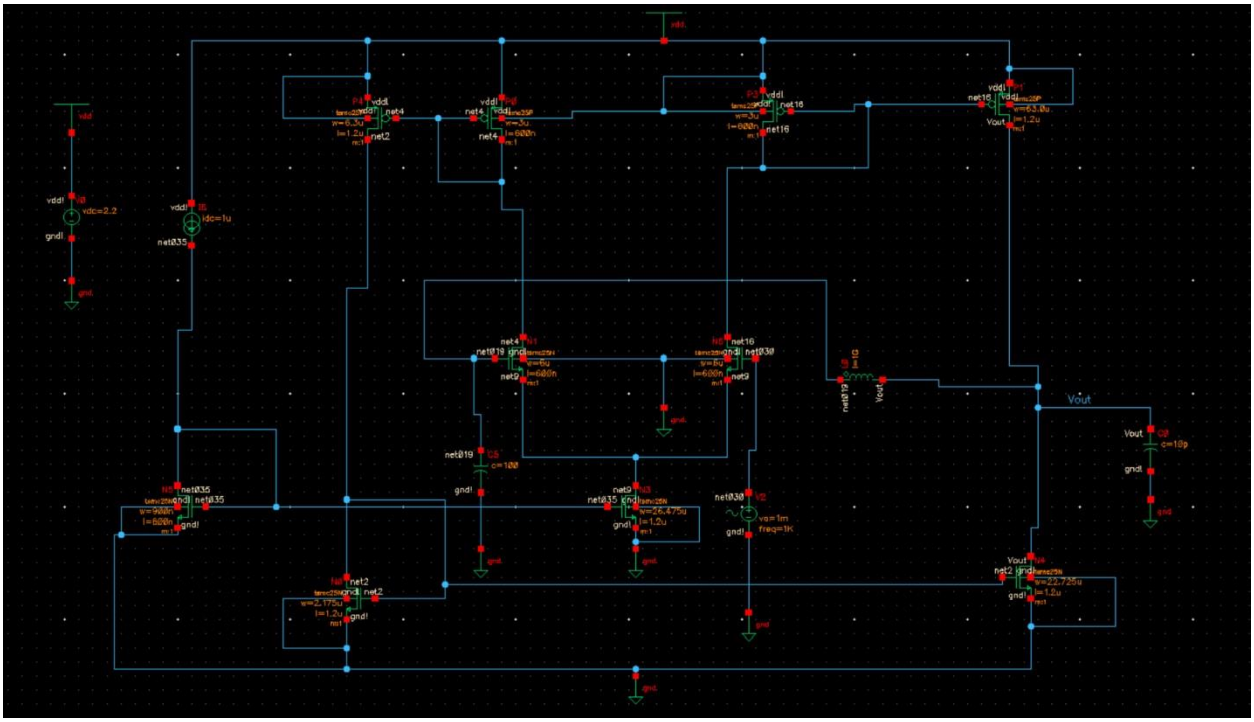
$M_3 = I = 10\,\mu A = \frac{1}{2} \times 0.9475 \times 10^{-4} \times \left(\frac{W}{L}\right) \times V_{ov}^2$

$\frac{W}{L} = 21.75 \Rightarrow L = 600\,nm$, $W = 13.05\,\mu m$

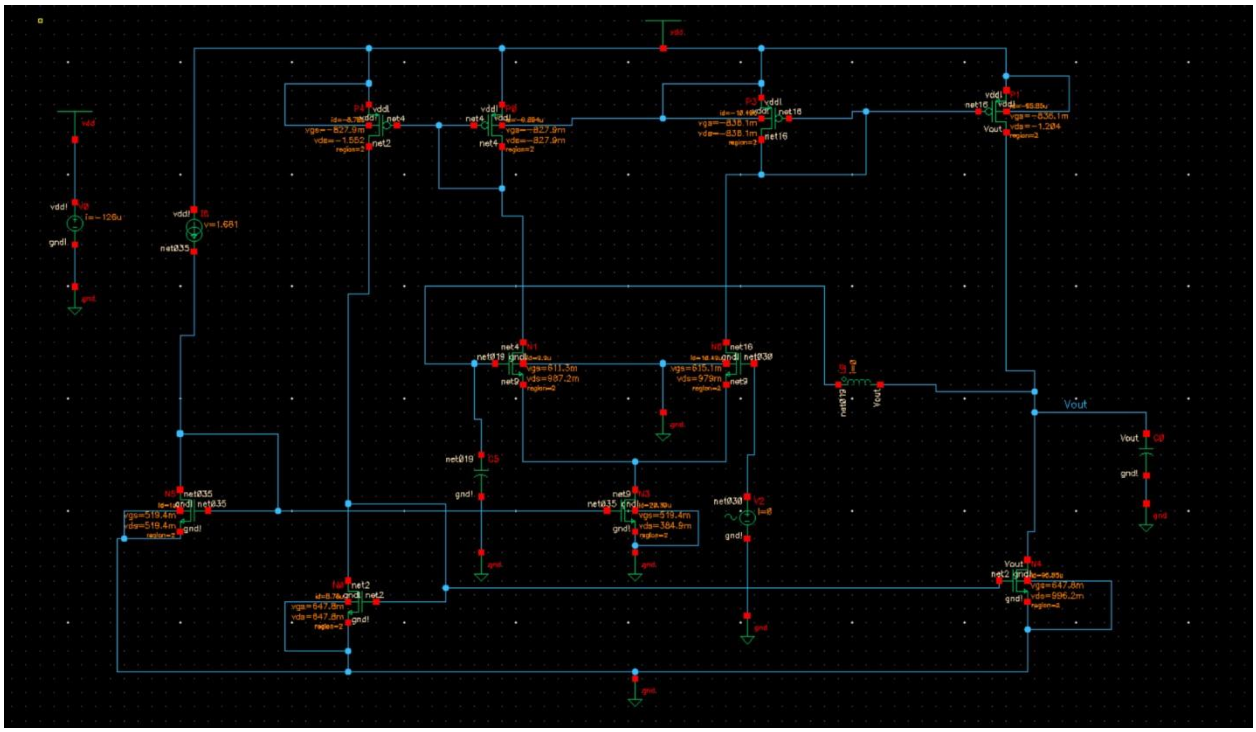
M_4 : $I = 100\,\mu A$; $W = 130.5\,\mu m$

a) Schematic:

- Voltage Gain >= 40 dB
- PDC < 0.3 mW
- BW > 200 KHz (while driving 10 PF load)



Amplifier schematic with DC operating points:



Power Consumption (PDC):

PDC = Id x Vdd = 126u X 2.2 = 0.27mW < 0.3 mW

b) AC Analysis:

Poles and zeros calculations

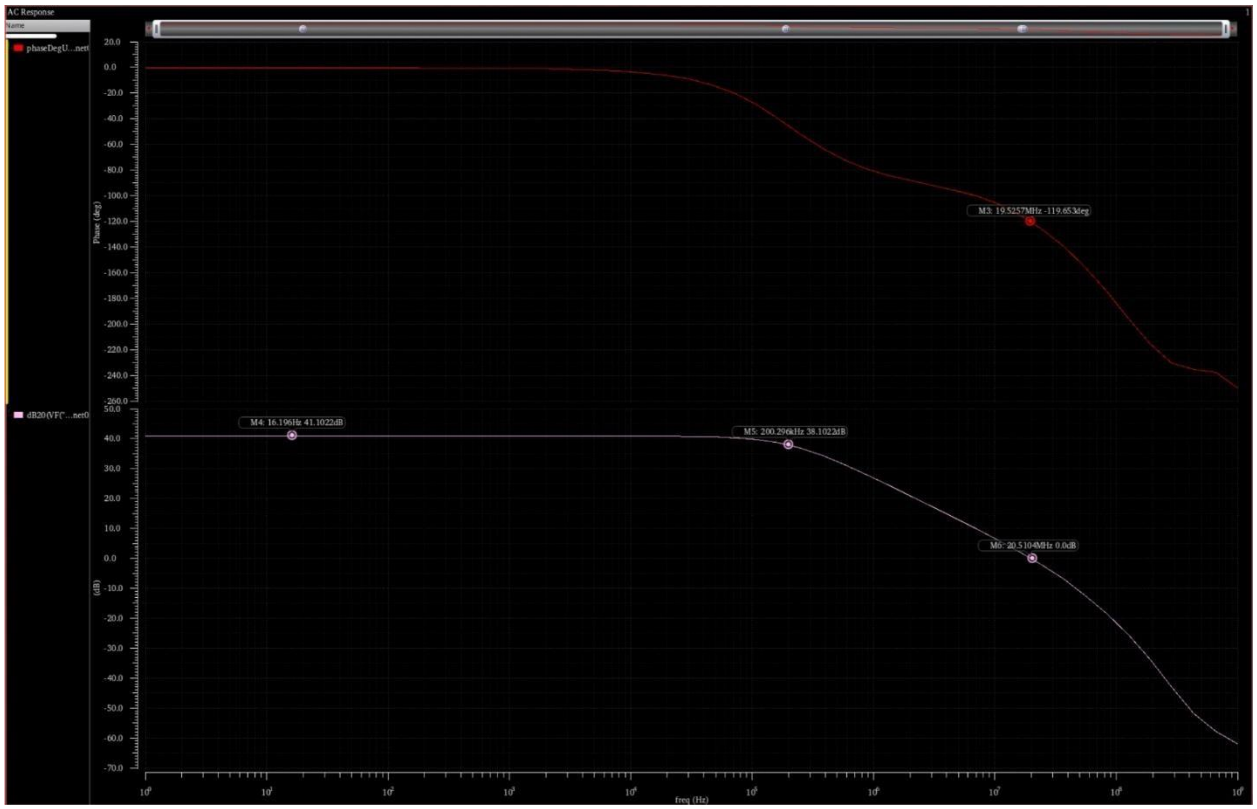
$$\text{Pole}: \quad r_o = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 100\mu} = 0.1\,M\Omega$$

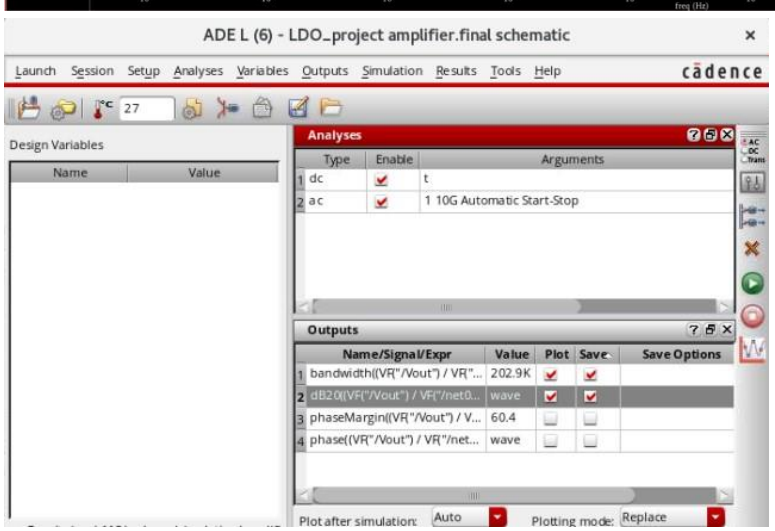
$$R_{out} = r_o \,//\, r_o = 0.05\,M\Omega$$

$$= 50\,K\Omega$$

$$\text{Pole} = \frac{1}{RC} = \frac{1}{50k \times 10p} = 2\,MHz = f_T$$

Frequency Response





C) Comparison of Calculations:

AC gain and bandwidth Calculations:

- The value of gain calculated by theoretical expressions was = 40dB, and the value obtained by simulation was around 41dB, which is > 40dB.
- The phase margin at 0dB gain was observed to be 60.4 degrees > 60degrees

• The power consumption PDC was observed to be 0.272mW < 0.3mW.

Gain BW hand calculations

Bandwidth : $f = \frac{1}{2\pi RC} = \frac{1}{2\pi (r_o \| r_o) C}$

$= \frac{1}{2\pi \times 50K \times 10pF} = 318\ KHz$

Gain of Common Source

$g_m = \frac{2 I_D}{V_{ov}} = \frac{2 \times 100\mu A}{0.1} = 2mS$

$r_o \Rightarrow \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 100 \times 10^{-6}} = 100K$

$A \Rightarrow g_m (r_o \| r_o)$

$\Rightarrow 2mS \times 50K$

$\Rightarrow 100\ V/V$

$\Rightarrow 40\ dB$

Gain of diff. amplifier

$$g_m = \frac{2I}{V_{ov}} = \frac{2 \times 12\,\mu A}{0.1} = 0.24$$

$$r_o = \frac{1}{\lambda I_D} = \frac{1}{0.1 \times 12\,\mu A} = 0.83\,M$$

$$A = g_m (r_o \| r_o) = 99.6\ V/V$$

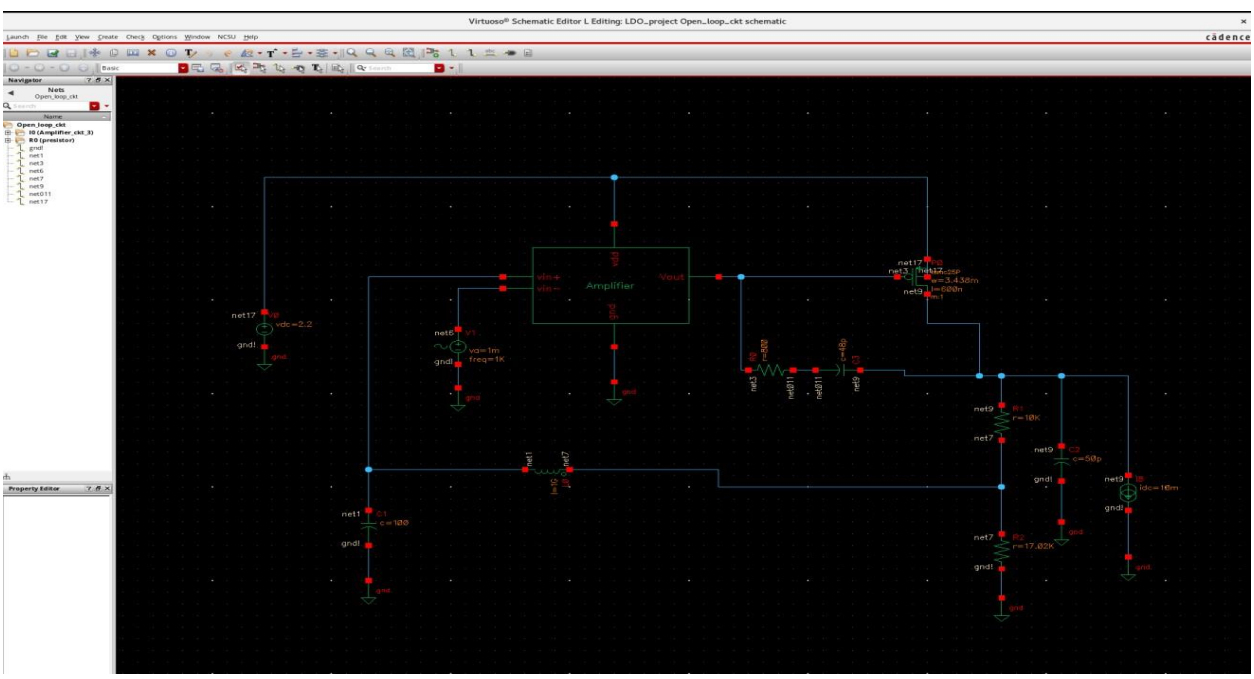
$$= 39.96\,dB \approx 40\,dB$$

$$\text{Total gain} = 40\,dB + 40\,dB$$

$$= 80\,dB$$

a) Open Loop:

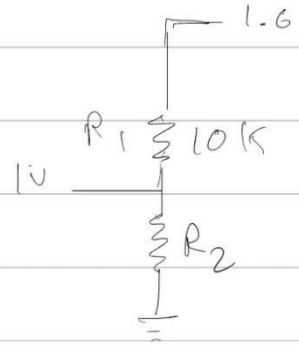
Schematic :



Calculations of R2 from the graph:

LDO

open loop

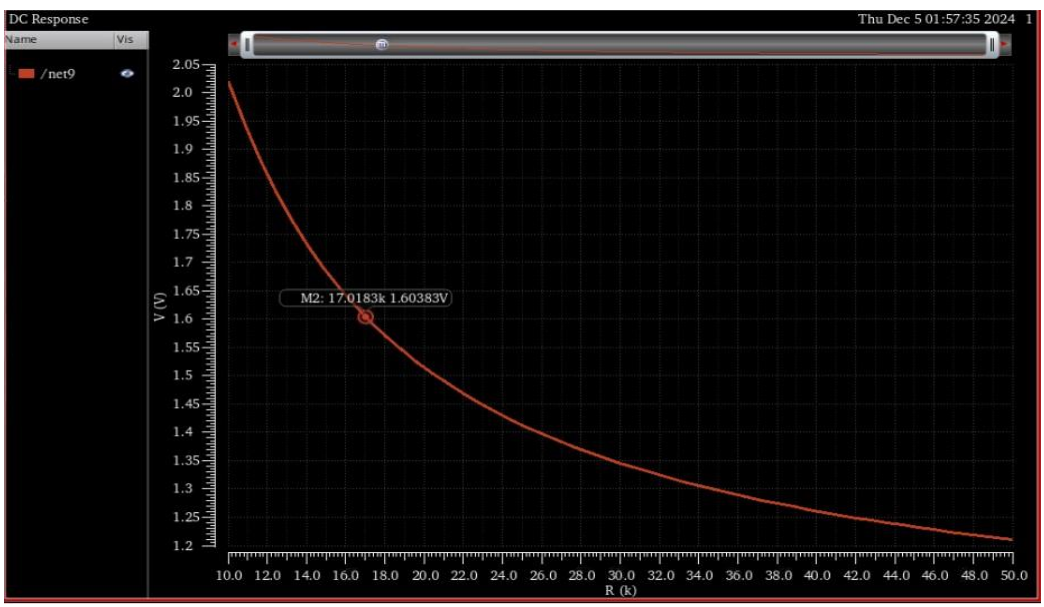


$$\frac{R_2}{R_1 + R_2}(1.6) = 1$$

$$\left(\frac{R_2}{10k + R_2}\right) 1.6 = 1$$

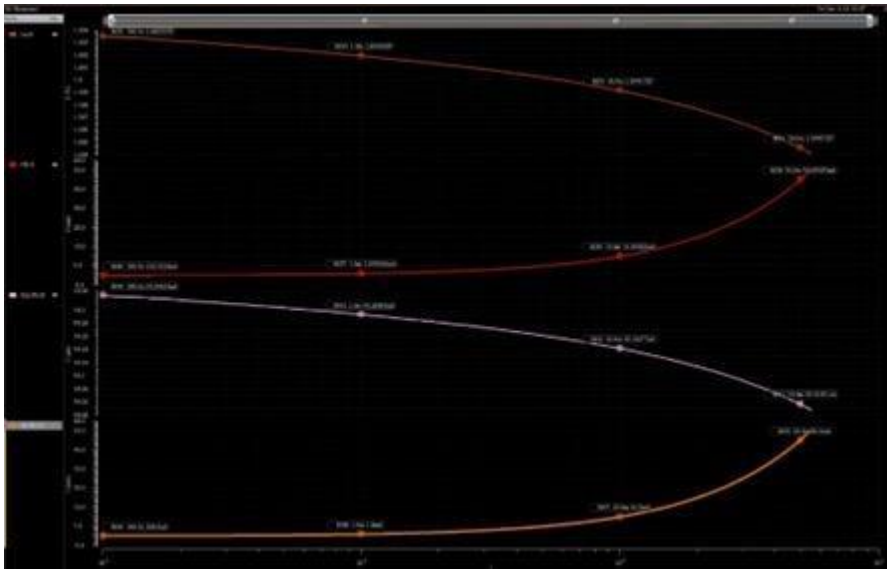
$$1.6 R_2 = 10K + R_2$$

$$0.6 R_2 = 10K \Rightarrow R_2 = 16.67 \, k\Omega$$

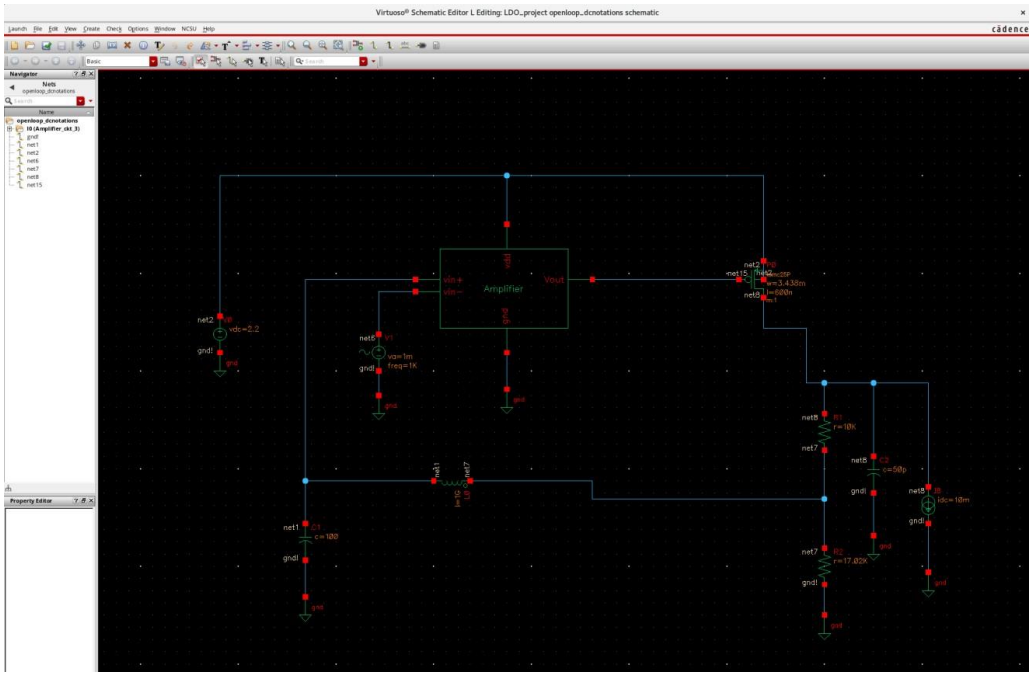


R2 = 17.018 K Ohms

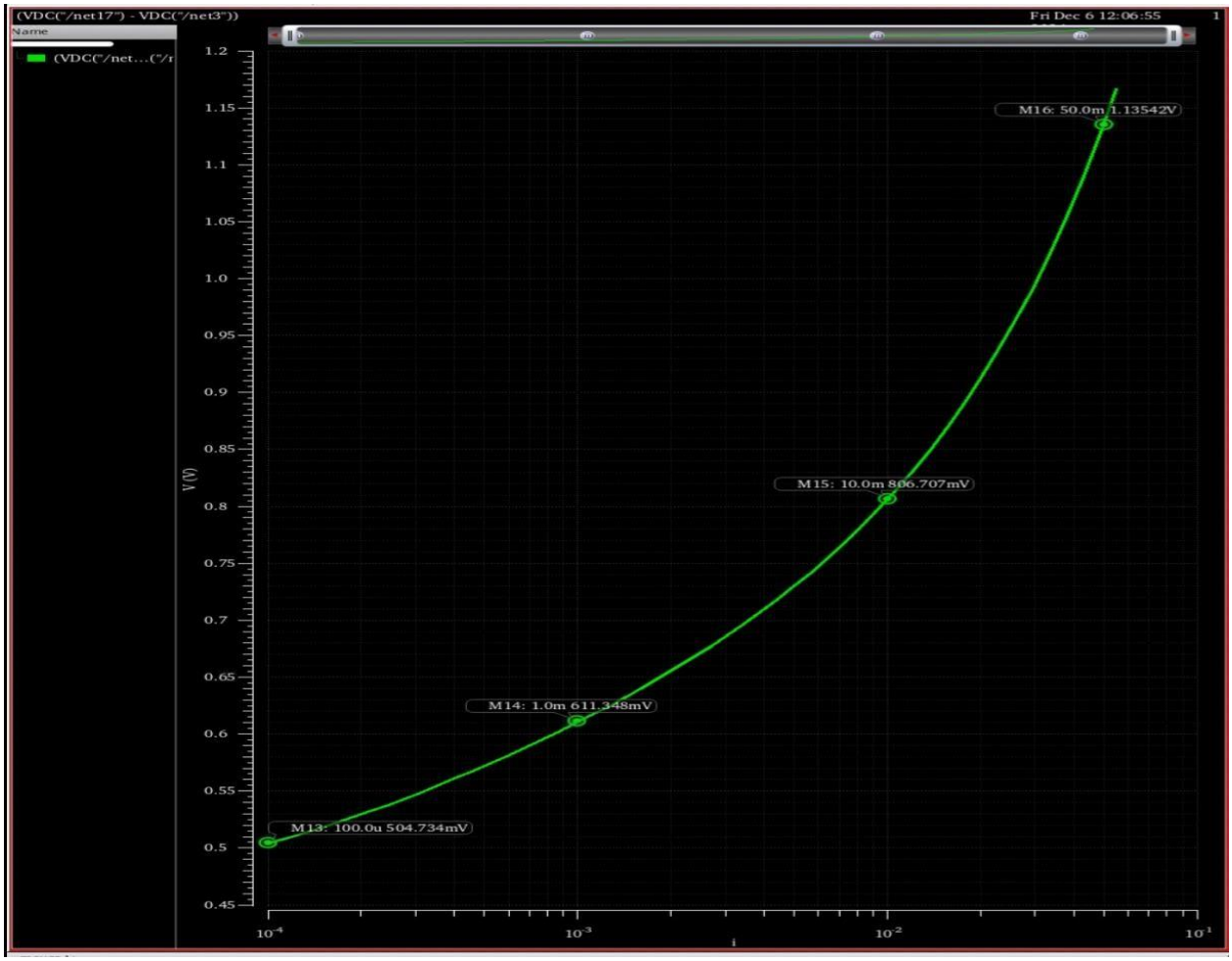
a) **DC Analysis**



DC operating points



Open Loop



Il	Vout (V)	Vsg	Isd	IRL(uA)
0.1mA	1.604	504.7m	-159.4u	59.85
1mA	1.602	611.4m	-1.059m	59.29
10mA	1.599	806.8m	-10.06m	59.19
50mA	1.595	1.136	-50.06m	59.01

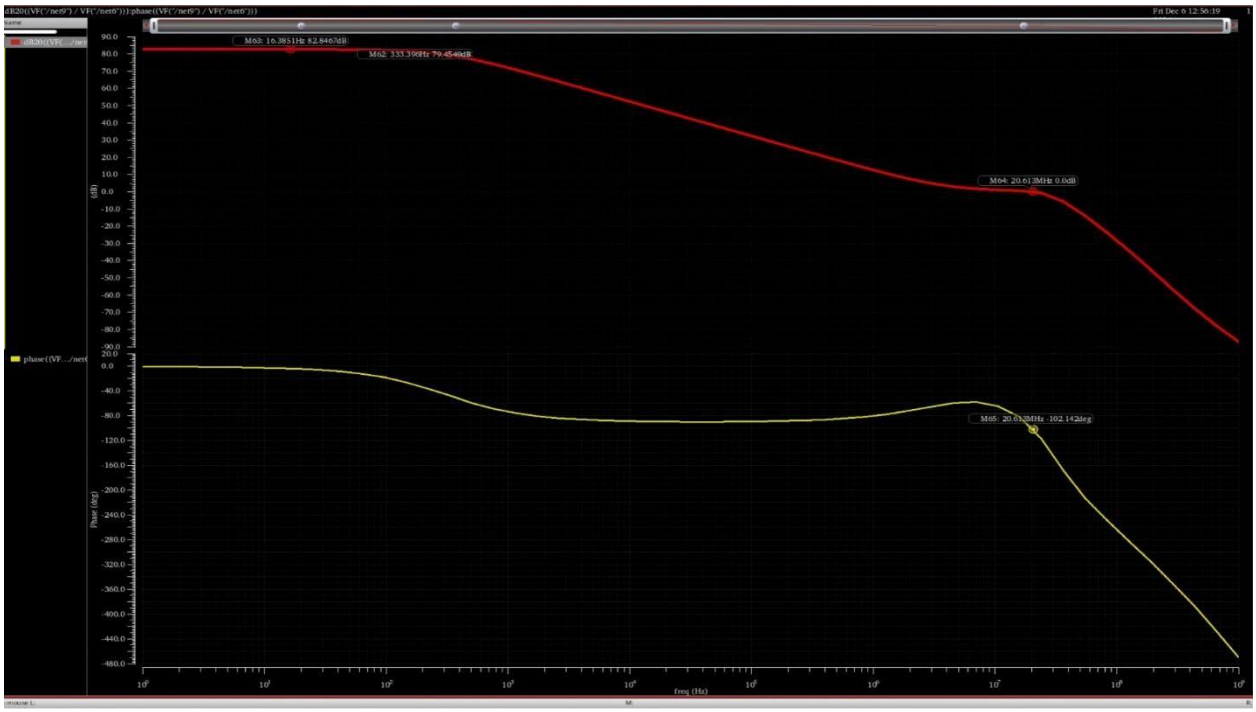
b) AC analysis:

Here Phase margin is lower than 60 degrees Hence,
Using frequency compensation:

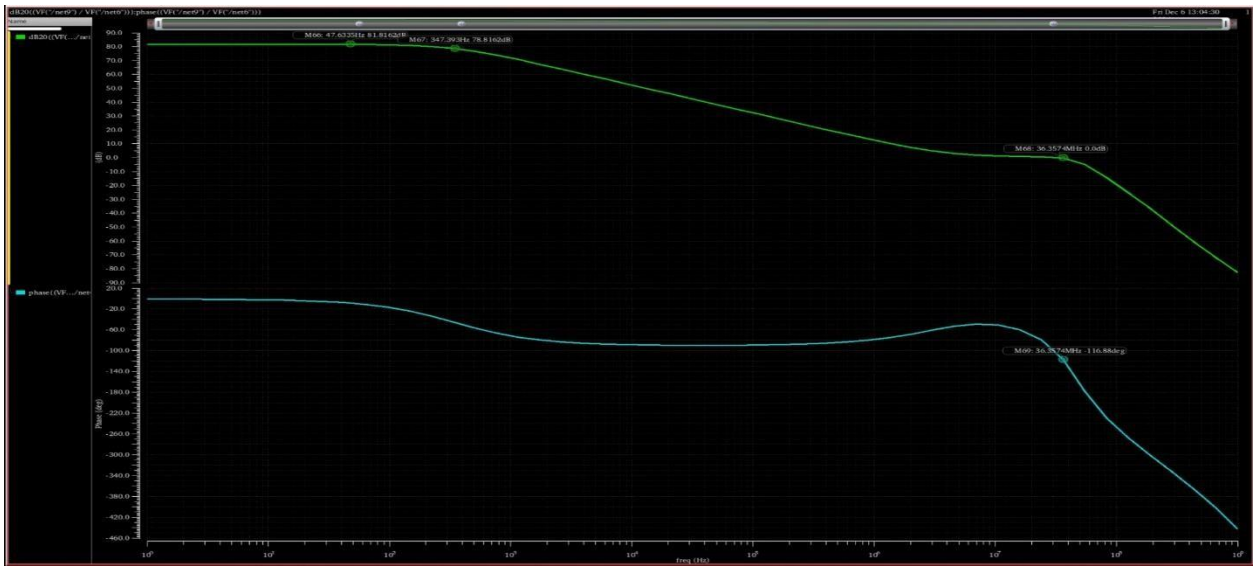
Values of R and C: **R= 800 ohms, C= 48pF**

Frequency Response for loads 1mA, 5mA, 10mA, 50mA With

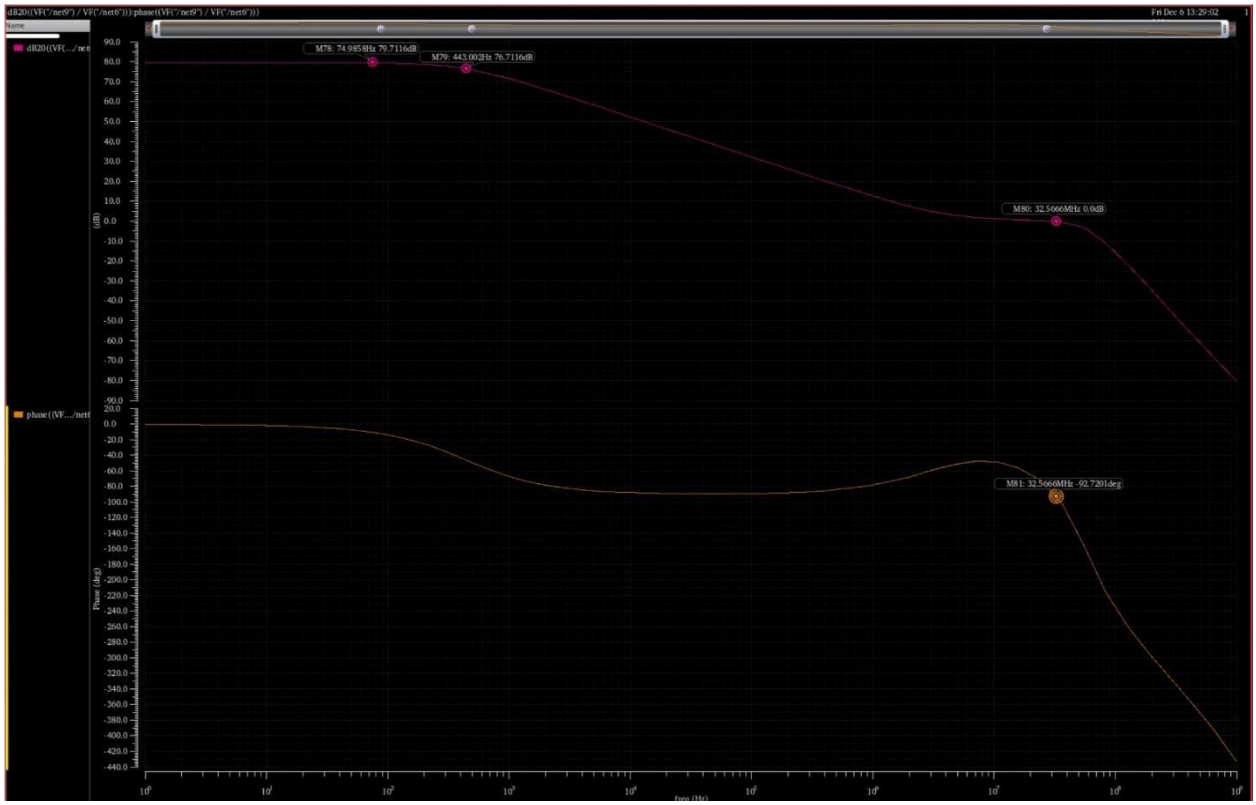
Feedback:



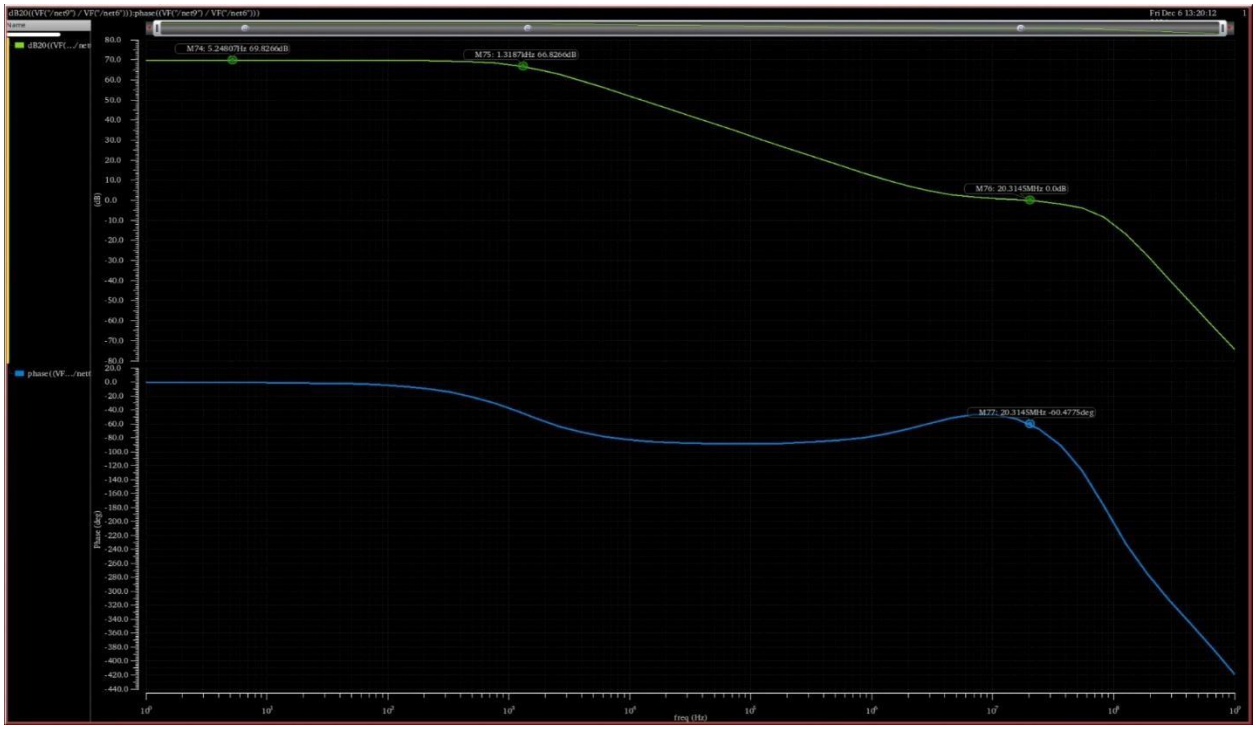
1mA



5mA


dB20((VF("/net9") / VF("/net6"))):phase((VF("/net9") / VF("/net6")))
Fri Dec 6 13:29:02
M78: 74.9858Hz 79.7116dB
M79: 443.002Hz 76.7116dB
M80: 32.5666MHz 0.0dB
M81: 32.5666MHz -92.7201deg
(dB)
Phase (deg)
freq (Hz)

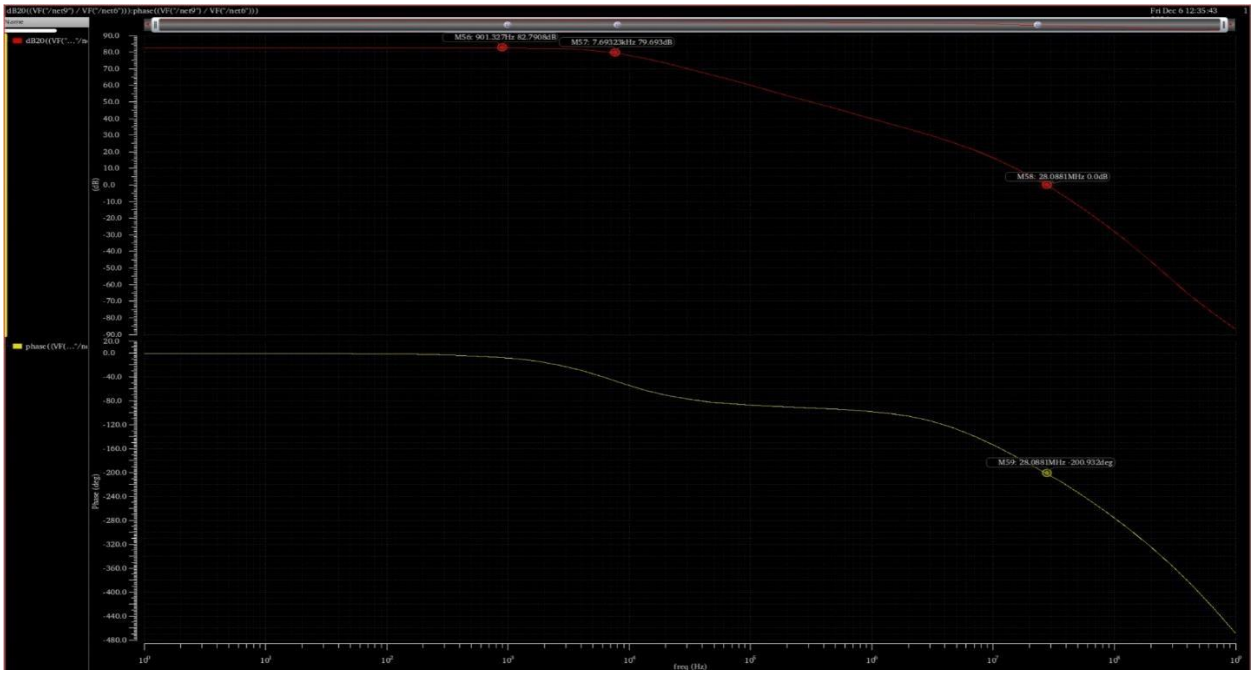
10mA


dB20((VF("/net9") / VF("/net6"))):phase((VF("/net9") / VF("/net6")))
Fri Dec 6 13:20:12
M74: 5.24807Hz 69.8266dB
M75: 1.3187kHz 66.8266dB
M76: 20.3145MHz 0.0dB
M77: 20.3145MHz -60.4775deg
(dB)
Phase (deg)
freq (Hz)

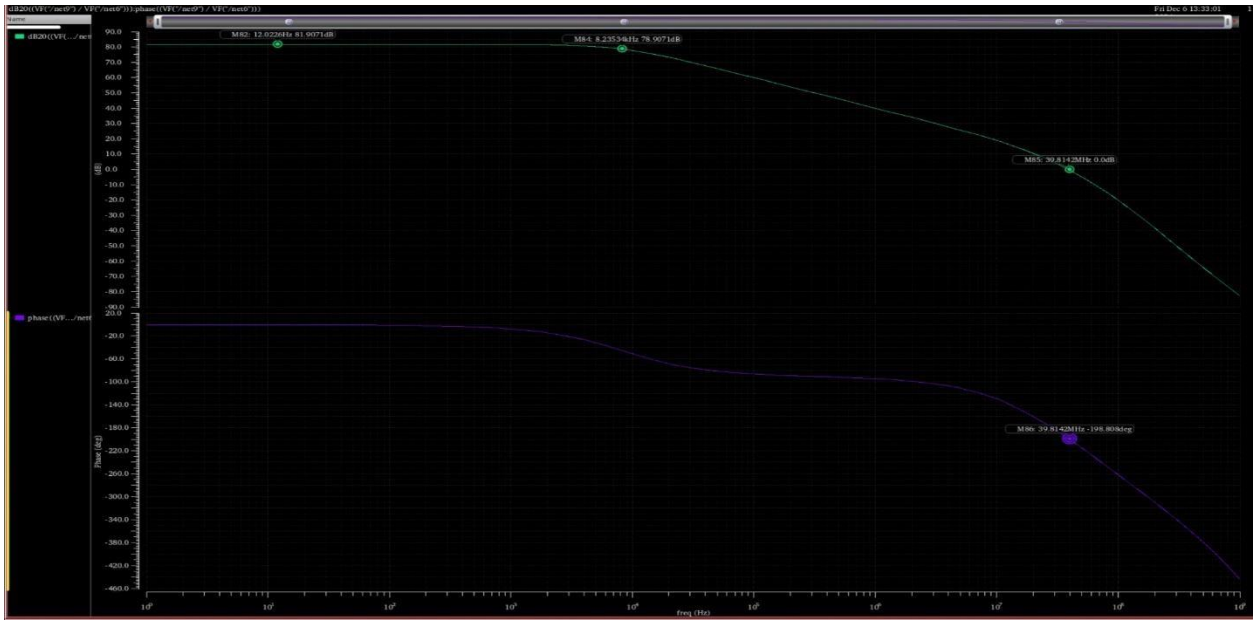
50mA

Frequency Response for loads 1mA, 5mA, 10mA, 50mA Without

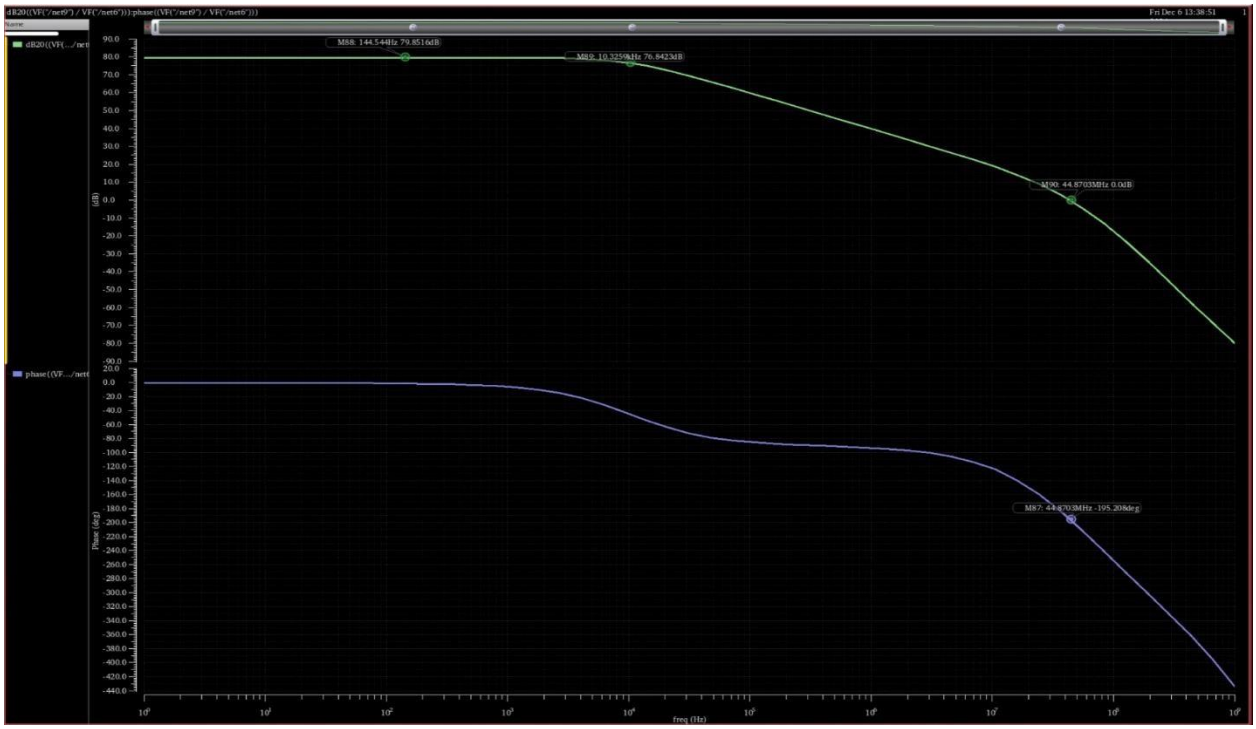
Feedback:



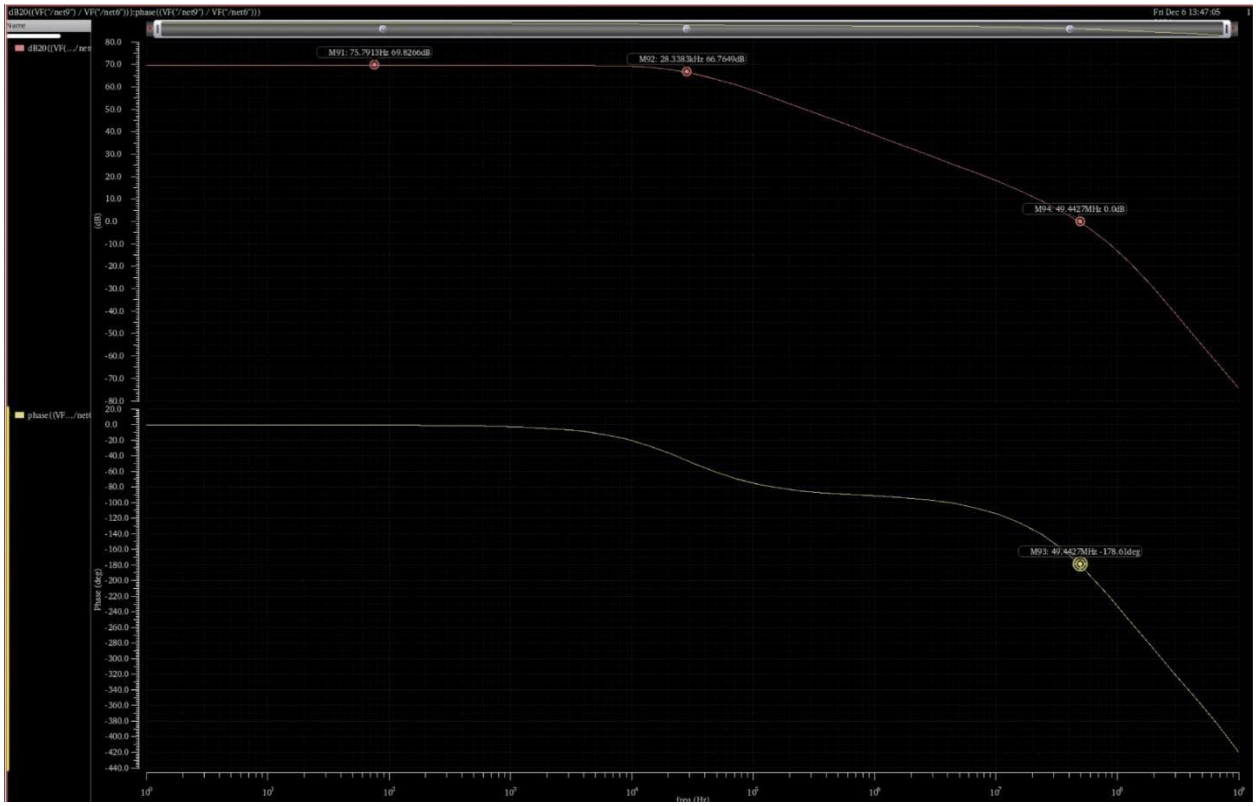
1mA



5mA

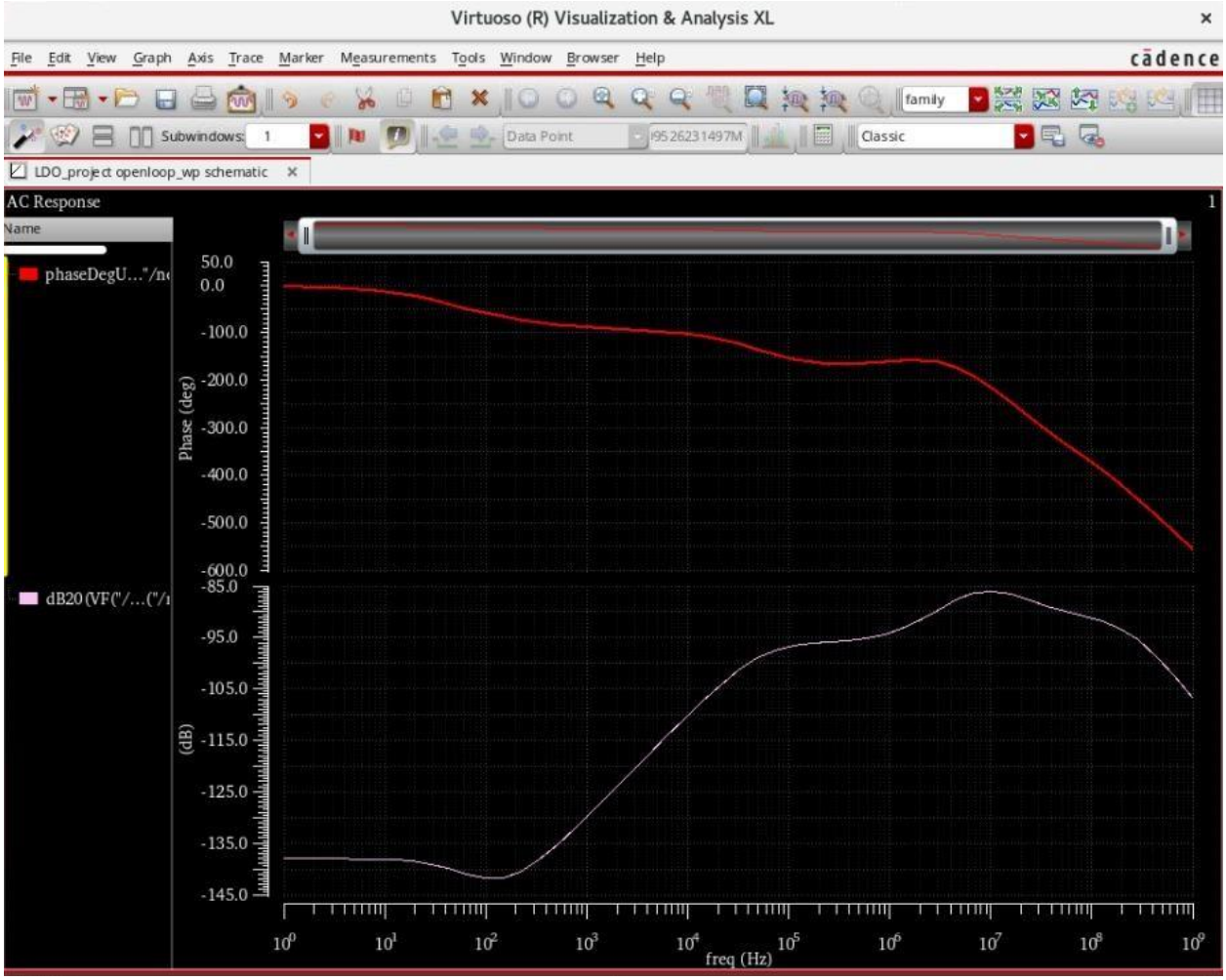

dB20((VF("/net9") / VF("/net6")));phase((VF("/net9") / VF("/net6")))
Fri Dec 6 13:38:51
M88: 144.544Hz 79.8516dB
M89: 10.3259kHz 76.8423dB
M90: 44.8703MHz 0.0dB
M87: 44.8703MHz -195.208deg
(dB)
Phase (deg)
freq (Hz)

10 mA


dB20((VF("/net9") / VF("/net6")));phase((VF("/net9") / VF("/net6")))
Fri Dec 6 13:47:05
M91: 75.7913Hz 69.8266dB
M92: 28.3383kHz 66.7649dB
M94: 49.4427MHz 0.0dB
M93: 49.4427MHz -178.61deg
(dB)
Phase (deg)
freq (Hz)

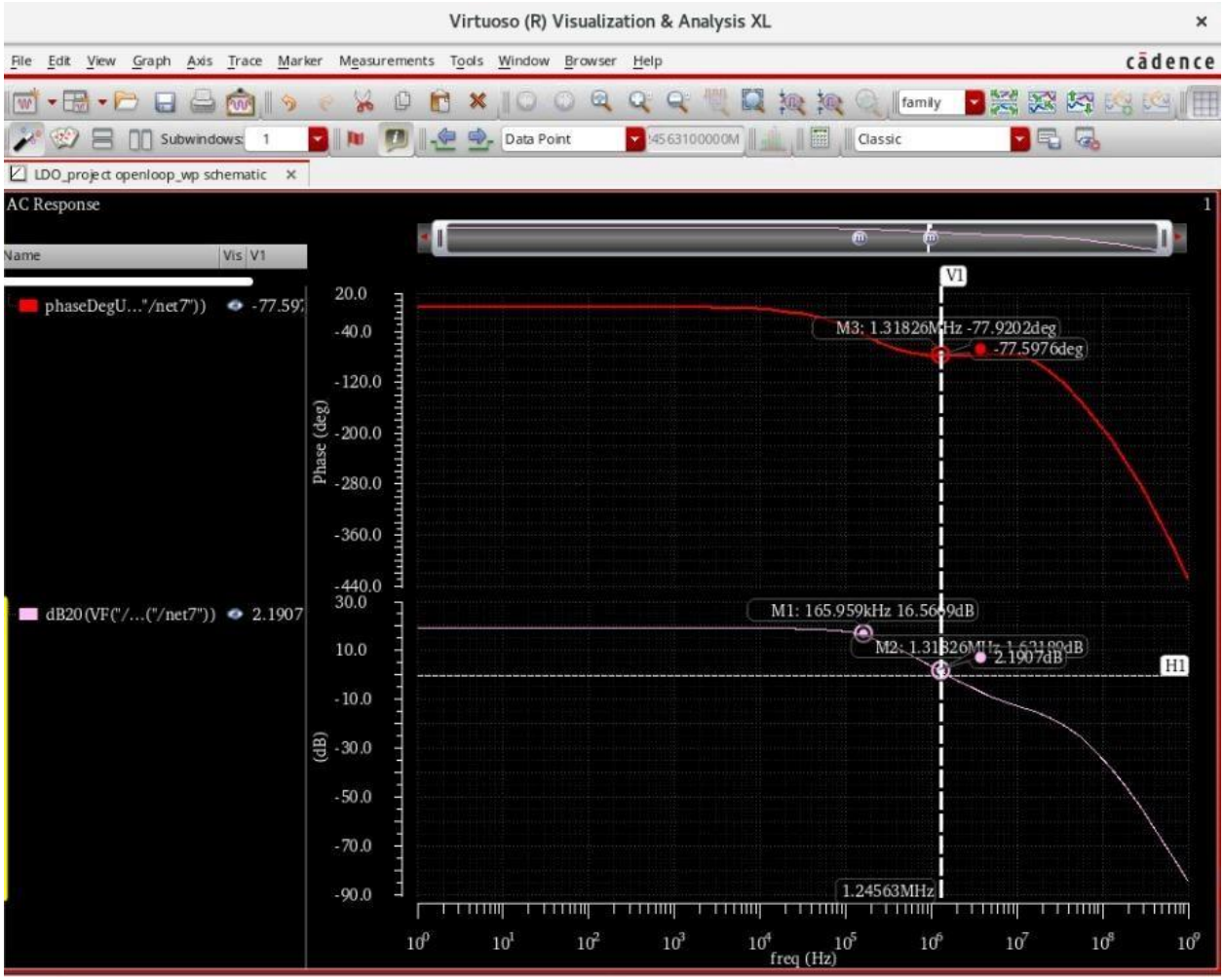
50 mA

Sweeping the width of the driver transistor (10 um, 100 um, 500 um, 1mm) and plotting the phase margin keeping the load current (iL) at a constant value of 5 mA and plotting various phase margin plots and tabulating the results.

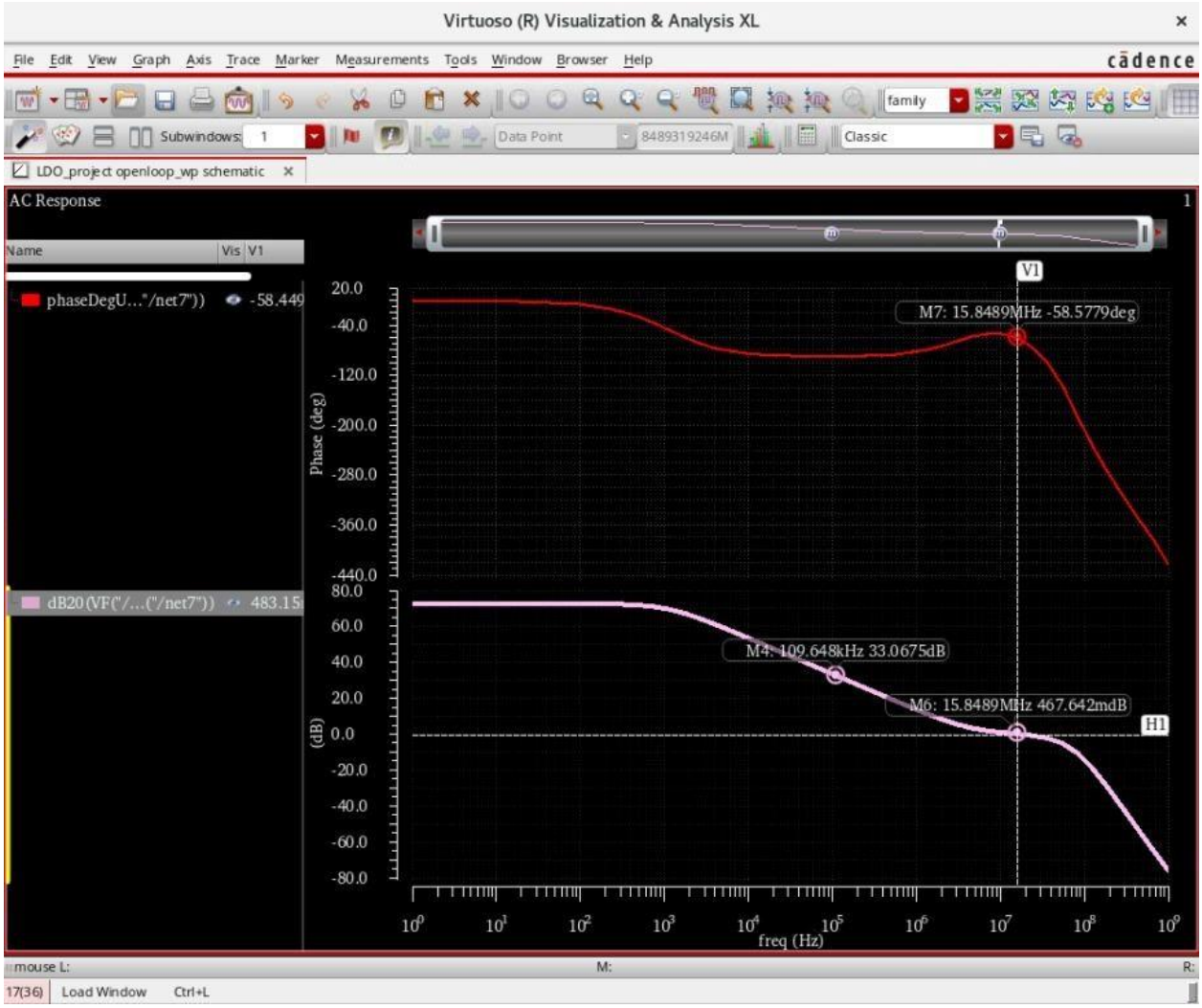


phase margin plots width 10um

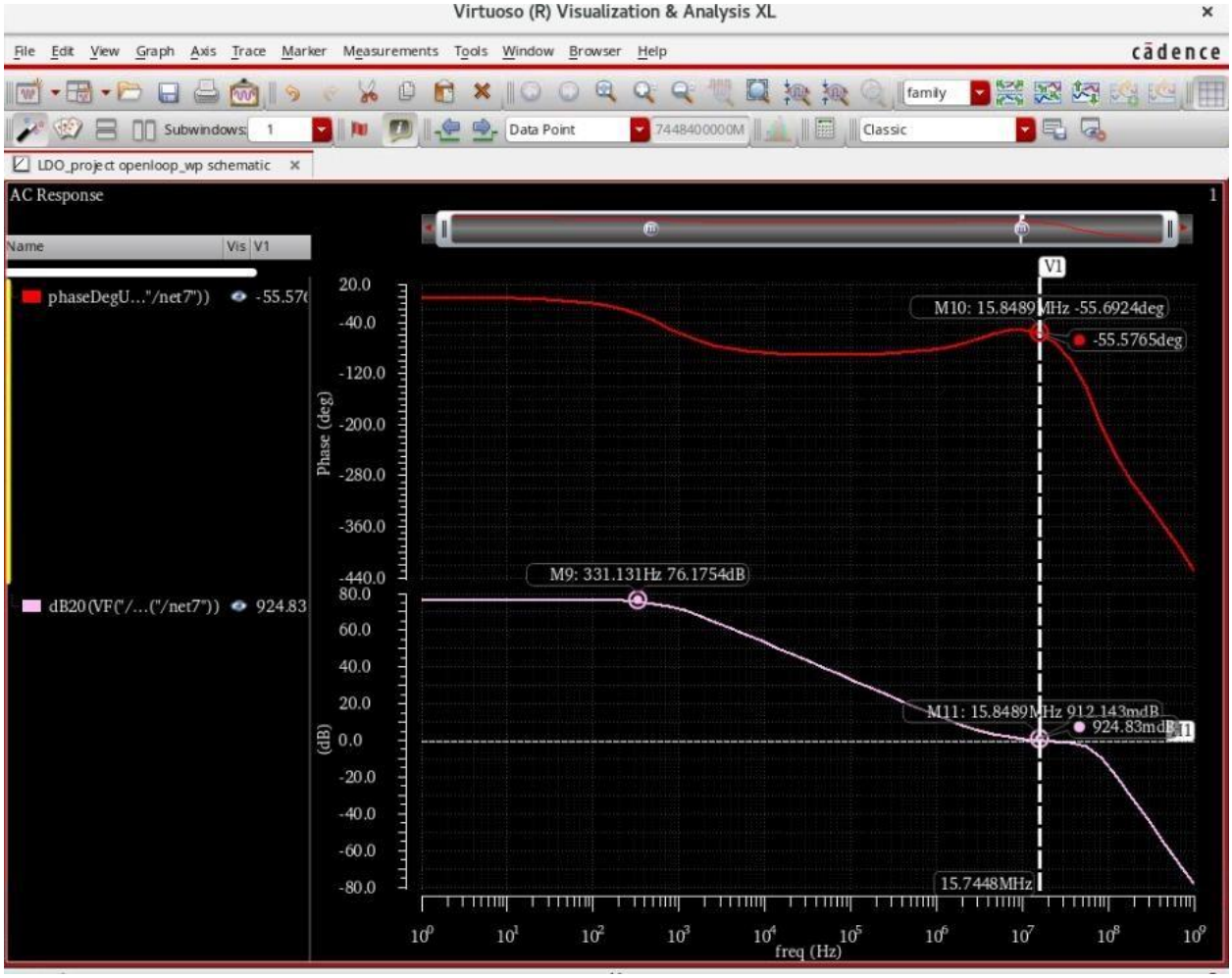
as the gain of the output is low the sweep at 10um is not correct for the driver transistor



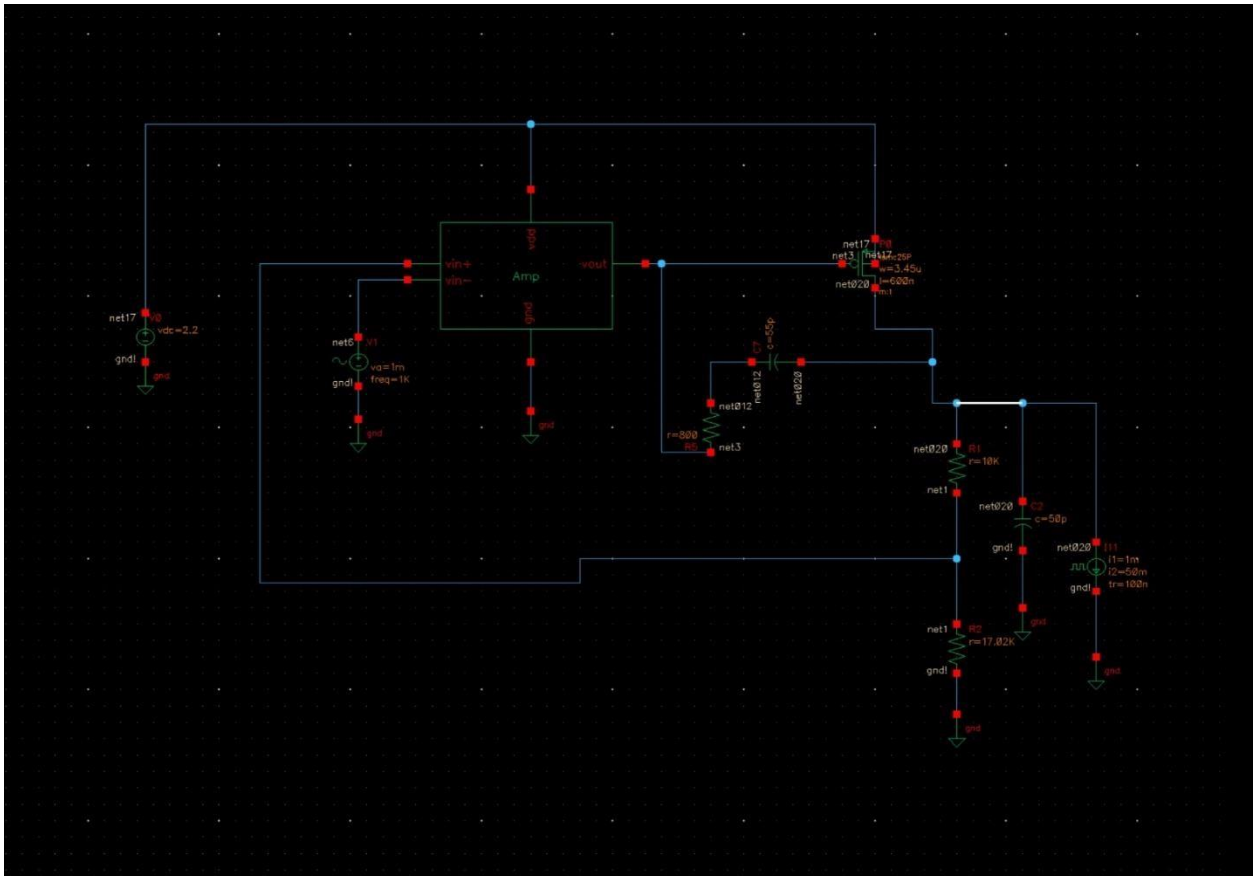
phase margin plots width 100um



phase margin plots width 500um

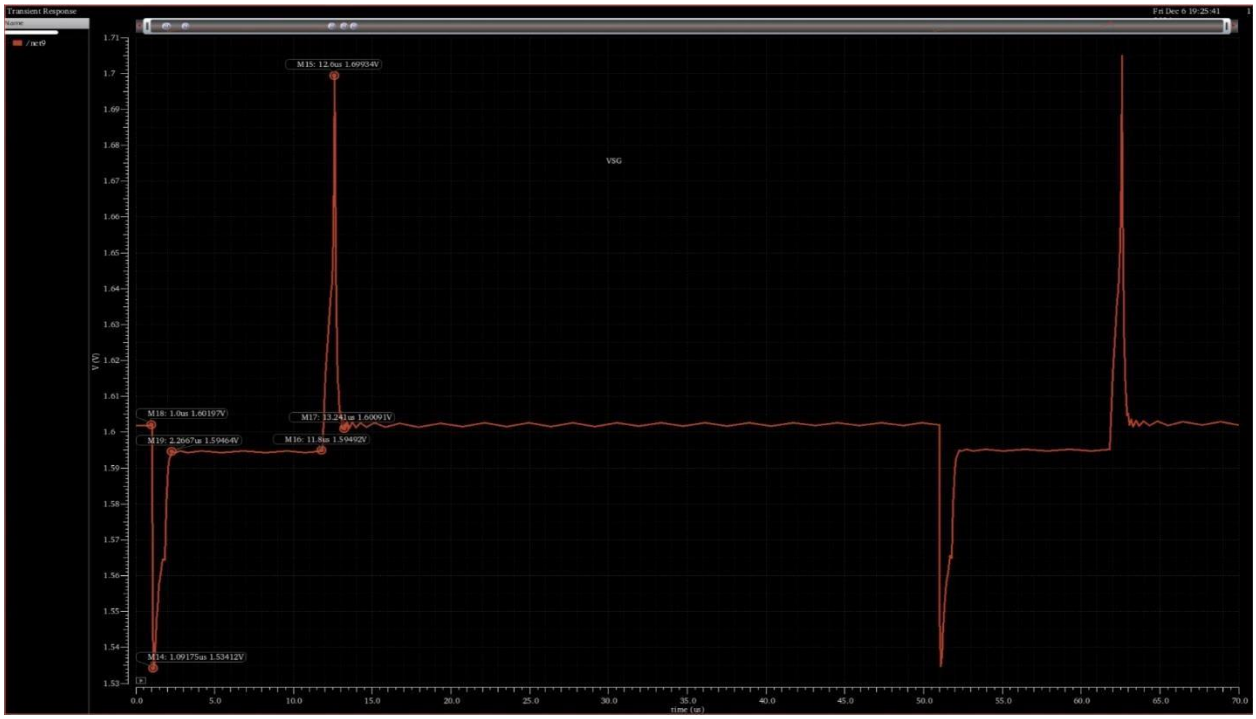


phase margin plots width 1mm

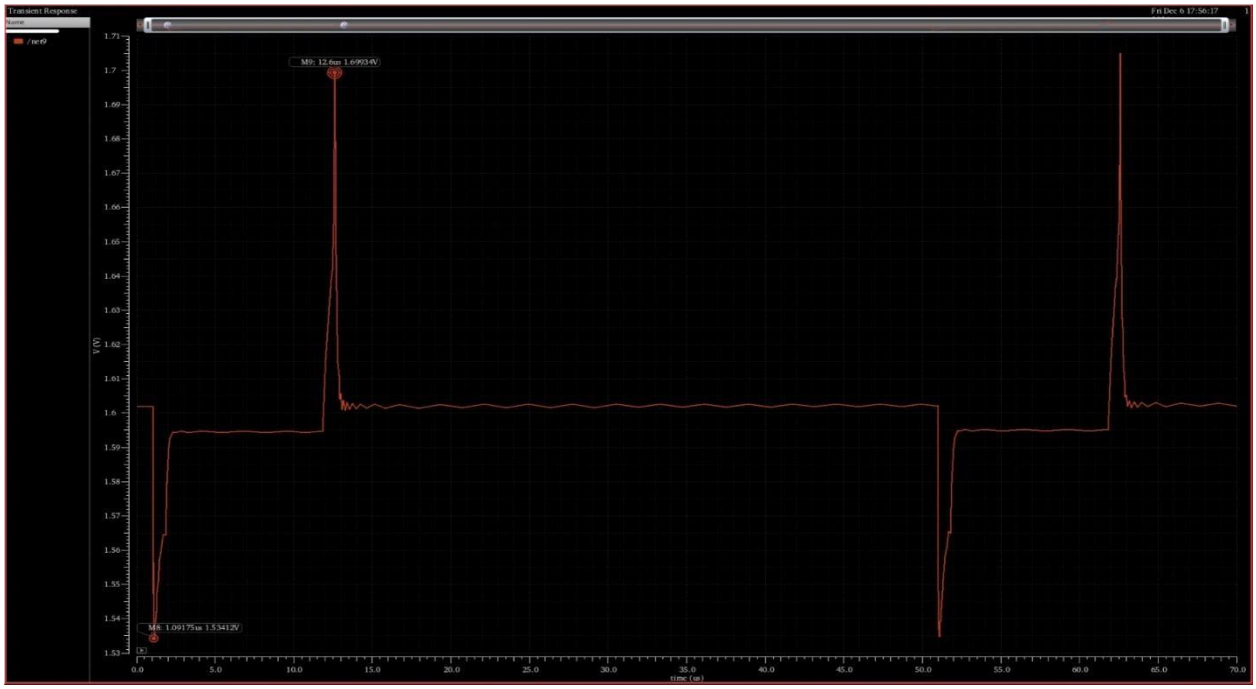


Closed Loop

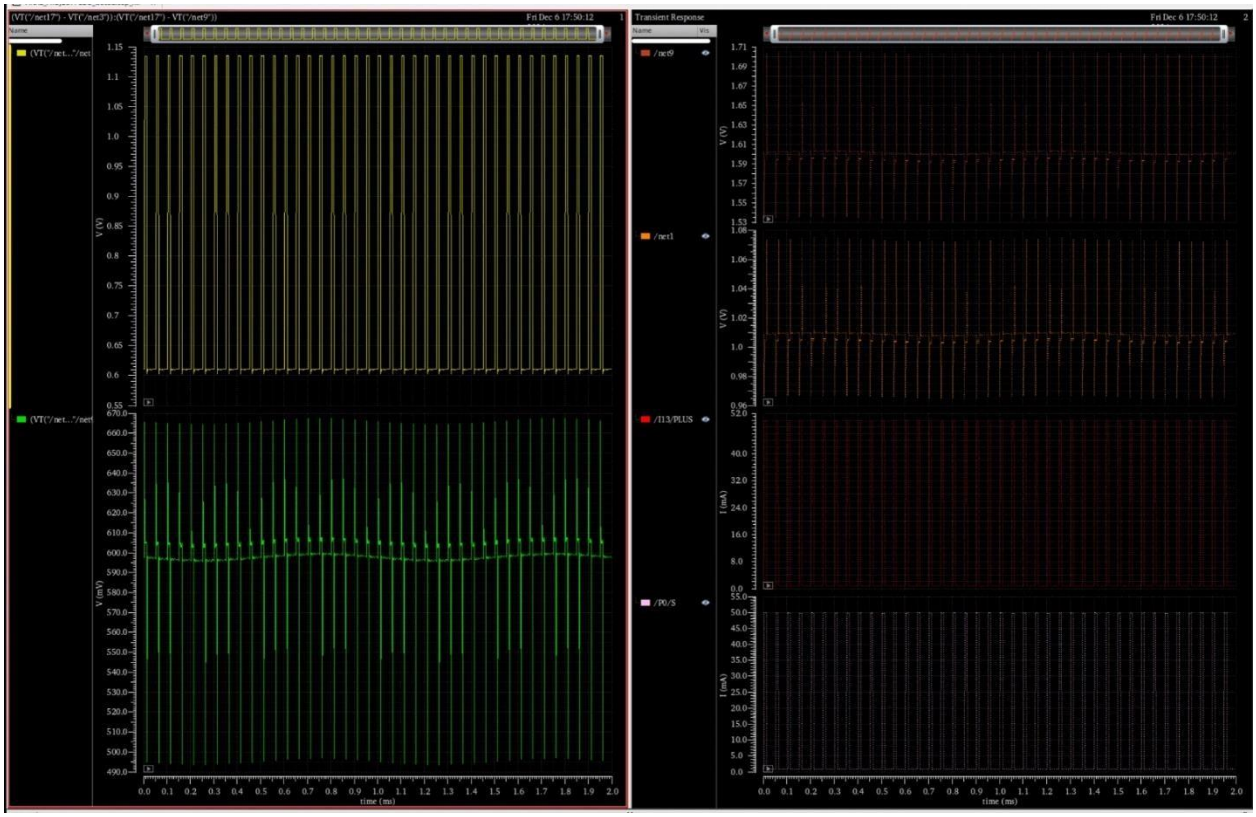
C) Transient analysis: (13 points)



ripple value and settling time in both low to high and high to low load current step change

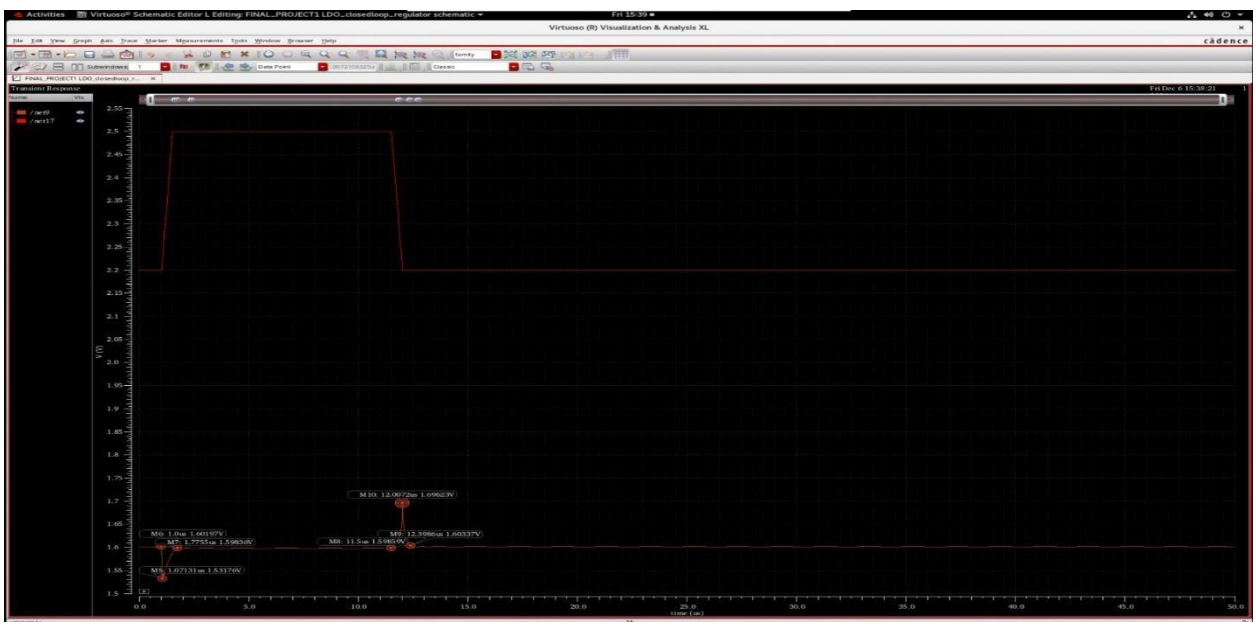


output voltage Vreg



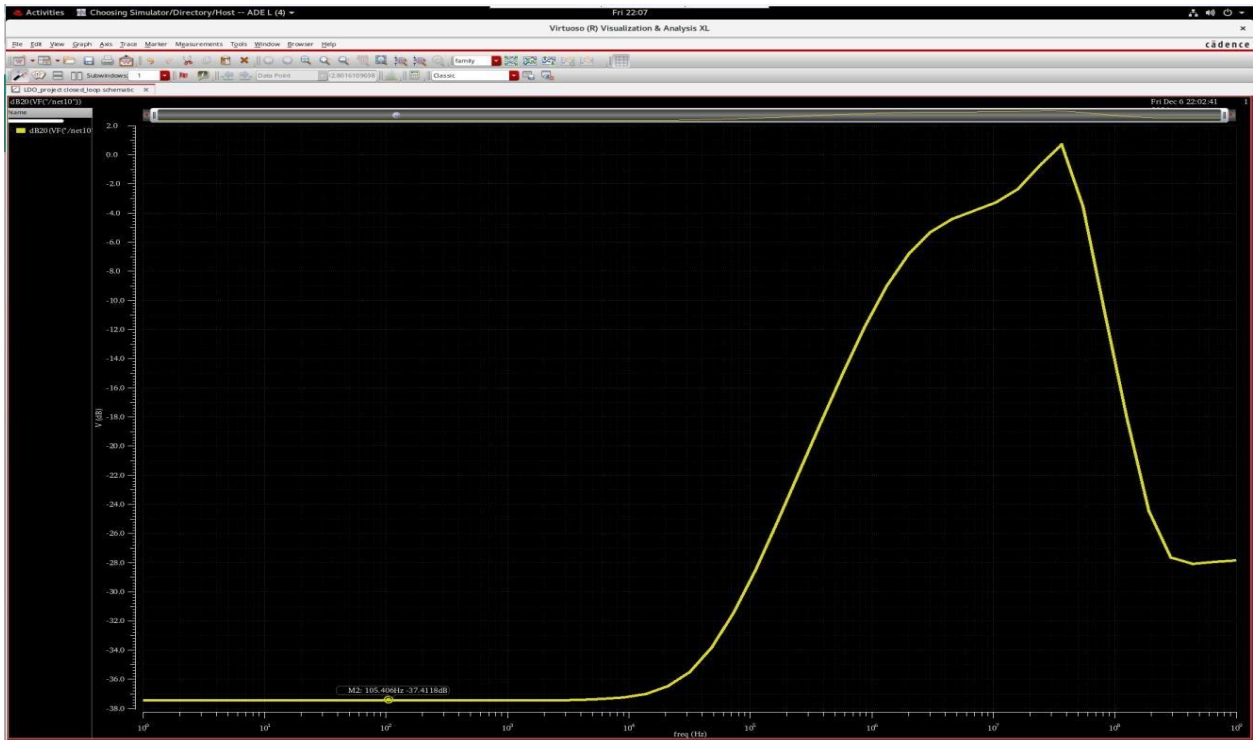
Plot IL, ISD, VSG, VSD, VF (feedback) vs. time.

c) Regulation:

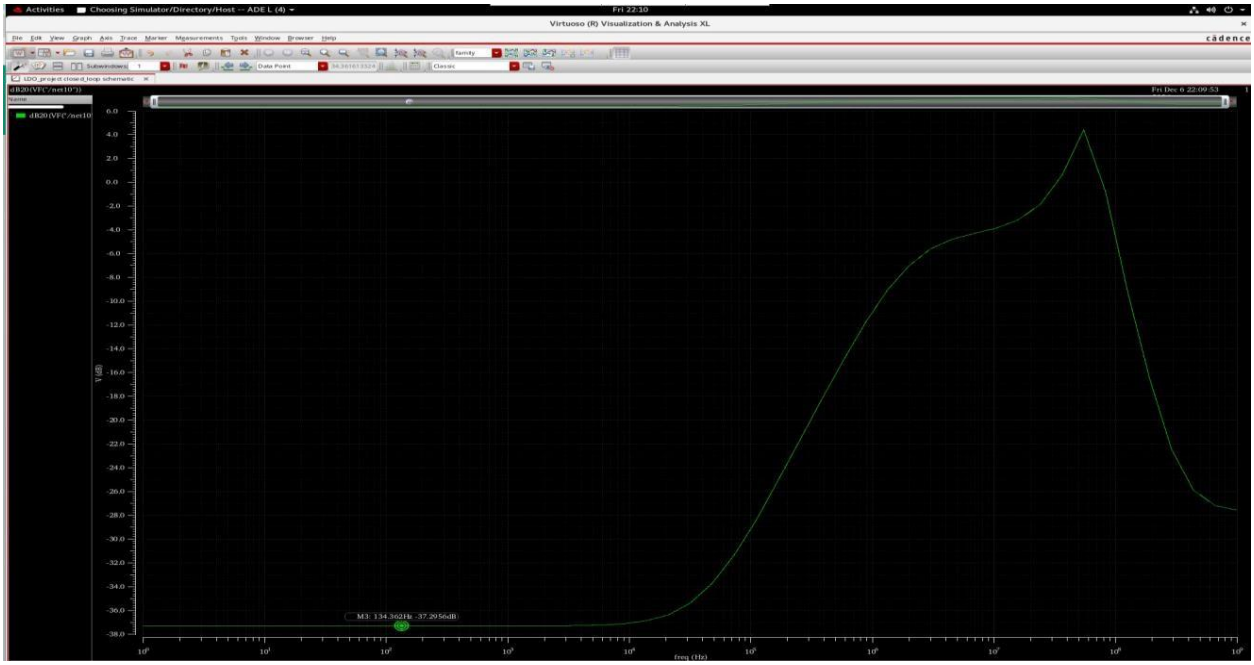


maximum voltage drop and settling time in both low-to-high and high-to-low supply step change

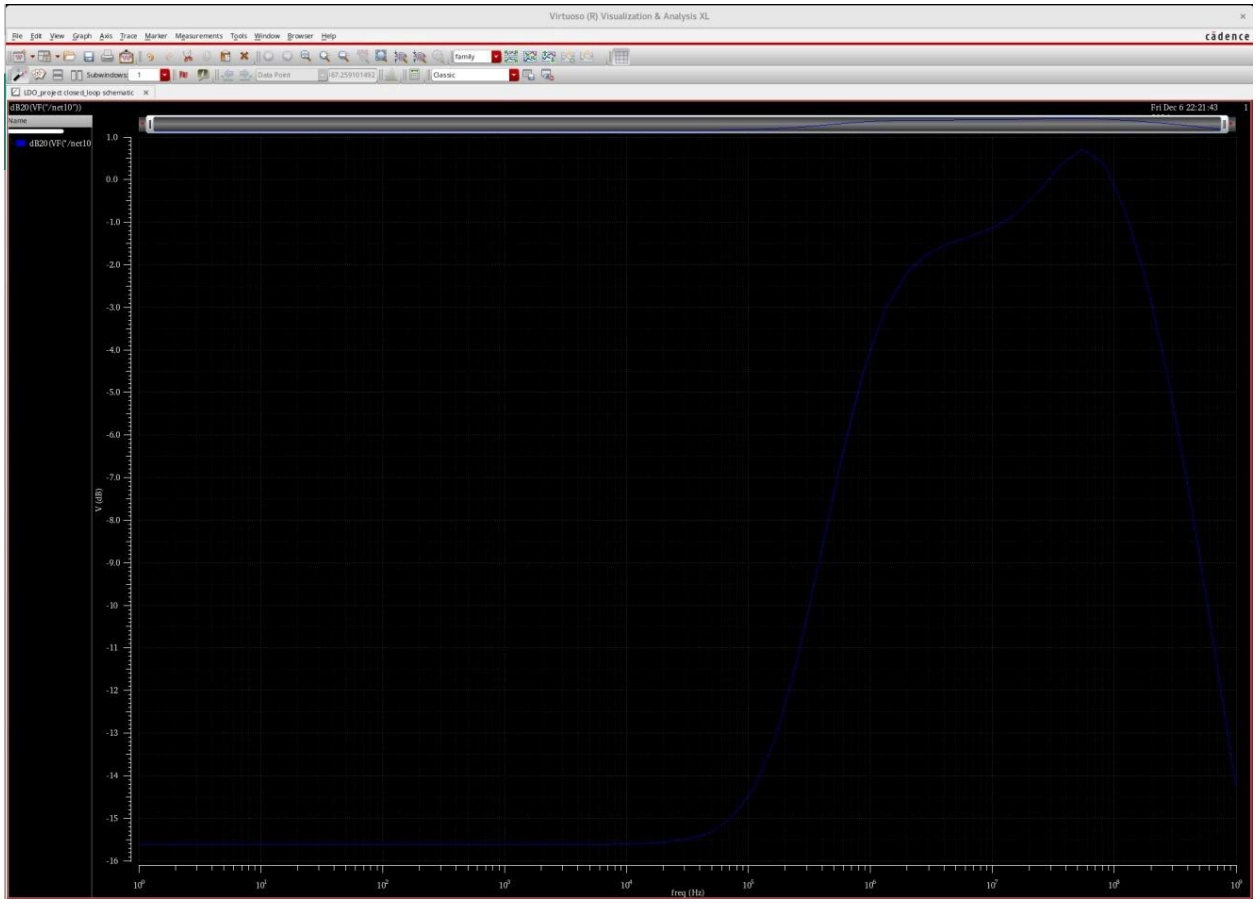
- **LDO output in dB at 1mA, 5 mA, 10 mA and 50 mA load condition**



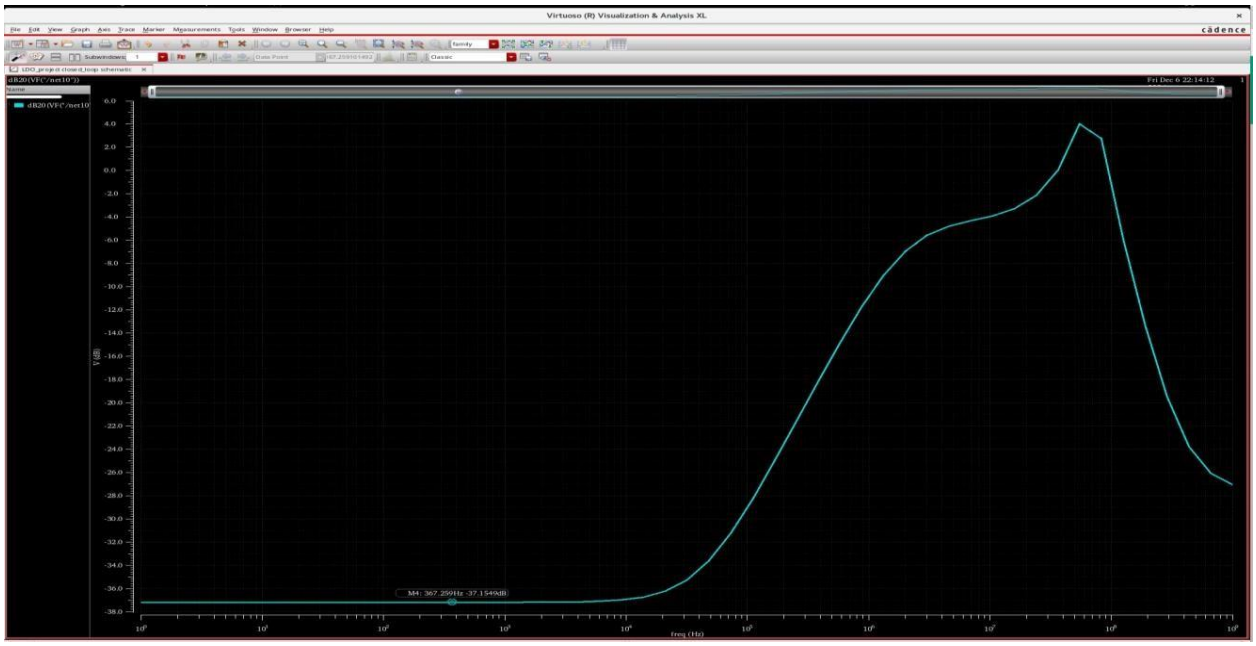
1 mA



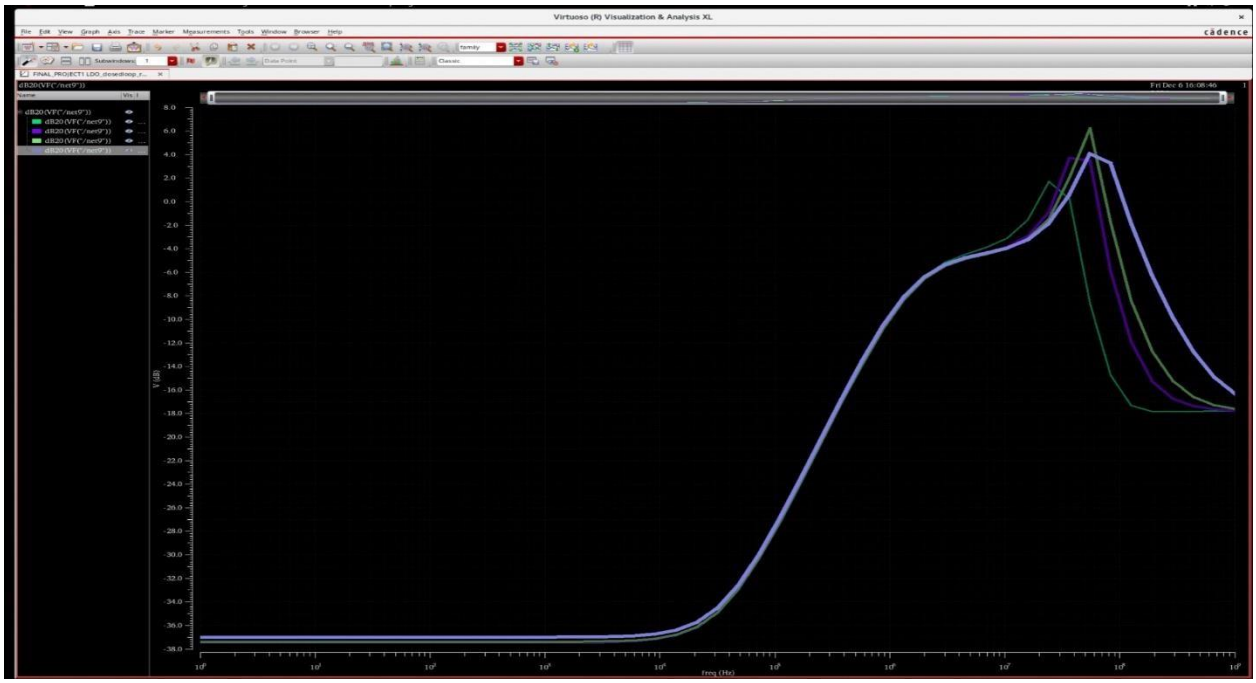
5mA


Virtuoso (R) Visualization & Analysis XL
cadence
dB20(VF("/net10"))
Fri Dec 6 22:21:43
V (dB)
freq (Hz)

10mA


Virtuoso (R) Visualization & Analysis XL
cadence
dB20(VF("/net10"))
Fri Dec 6 22:14:12
V (dB)
freq (Hz)
M4: 367.259Hz -37.1549dB

50mA



LDO output in dB at 1mA, 5 mA, 10 mA, and 50 mA load condition

To obtain the final output of LDO we are using parametric sweep

Conclusion:

To sum up, the low-dropout regulator (LDO) design, simulations, and analysis showed how to use an amplifier as a fundamental circuit for voltage regulation.

- By applying effective power management design techniques, LDO performance and dependability were increased.
- All of the project's requirements were fulfilled, and when the observed results were compared to the simulation and manually computed values, they appeared to be almost identical.